

Common Urinary Disease

Urinary tract infection (UTI)

1. In health bacterial colonisation is confined to the lower end of the urethra and the remainder of the urinary tract is sterile.
2. UTI is the most common bacterial infection managed in general medical practice and accounts for 1-3% of consultation.
3. Up to 50% of women have a UTI at some time.
4. The prevalence of UTI in women is about 3% at the age of 20, increasing by about 1% in each subsequent decade.
5. In male UTI is uncommon, except in the first year of life and in men over 60, in whom urinary tract obstruction due to prostatic hypertrophy may occur.

Aetiology and risk factors

1. Urine is an excellent culture medium for bacteria; in addition, the urothelium of susceptible persons may have more receptors to which virulent strains of *Escherichia coli* become adherent.
2. In women, the ascent of organ into the bladder is easier than in men; the urethra is shorter and absence of bactericidal prostatic secretions may be relevant.
3. Sexual intercourse may cause minor urethral trauma and transfer bacteria from the perineum into the bladder.
4. Instrumentation of the bladder may also introduce organisms.
5. Multiplication of organisms depends on a number of factors, including the size of the inoculum and virulence of the bacteria.

Conditions which predispose to UTI /Risk factors for urinary tract infection

Incomplete bladder emptying :

- Bladder outflow obstruction

- Neurological problems (e.g. multiple sclerosis, diabetic neuropathy)
- Gynaecological abnormalities (e.g. uterine prolapse)
- Vesico-ureteric reflux

Foreign bodies :

- Urethral catheter or ureteric stent

Loss of host defences

- Atrophic urethritis and vaginitis in post menopausal women
- Diabetes mellitus

Clinical assessment

Typical features of cystitis and urethritis include:

- Abrupt onset of frequency of micturition and urgency
- Scalding pain in the urethra during micturition (dysuria)
- Suprapubic pain during and after voiding
- Intense desire to pass more urine after micturition, due to spasm of the inflamed bladder wall (strangury)
- Urine that may appear cloudy and have an unpleasant odour
- Microscopic or visible haematuria.

Systemic symptoms are usually slight or absent

1. However, infection in the lower urinary tract can spread; acute pyelonephritis is suggested by prominent systemic symptoms with fever, rigors, vomiting, hypotension and loin pain, guarding or tenderness, and may be an indication for hospitalisation.
2. Only about 30% of patients with acute pyelonephritis have associated symptoms of cystitis or urethritis.
3. Prostatitis is suggested by systemic symptoms and prostatic tenderness on rectal examination.

- The spectrum of presentations of urinary tract infection
- Asymptomatic bacteriuria
- Symptomatic acute urethritis and cystitis
- Acute pyelonephritis
- Acute prostatitis
- Septicaemia (usually Gram-negative bacteria)

Differential diagnosis

1. The differential diagnosis of lower urinary tract symptoms includes urethritis due to sexually transmitted disease, notably chlamydia or Reiter's syndrome.
2. Some patients, usually female, have symptoms suggestive of urethritis and cystitis but no bacteria are cultured from the urine (the 'urethral syndrome').
3. Possible explanations include infection with organisms not readily cultured by ordinary methods (e.g. *Chlamydia*, certain anaerobes), intermittent or low-count bacteriuria, reaction to toiletries or disinfectants, symptoms related to sexual intercourse, or post-menopausal atrophic vaginitis.
4. The differential diagnosis of acute pyelonephritis includes acute appendicitis, diverticulitis, cholecystitis, salpingitis, ruptured ovarian cyst or ectopic pregnancy.
5. In perinephric abscess, there is marked pain and tenderness, and often bulging of the loin on the affected side.
6. Patients are extremely ill, with fever, leucocytosis and positive blood cultures, but urinary symptoms are absent and urine contains neither pus cells nor organisms.

Investigations urine

1. In an otherwise healthy woman with a single lower urinary tract infection, urine culture prior to treatment is not mandatory.
2. It is necessary, however, in patients with recurrent infection or after failure of initial treatment, during pregnancy, or in patients susceptible to serious infection (e.g. the

immunocompromised, those with diabetes or an indwelling catheter, or older people

3. Definitive diagnosis rests on the combination of typical clinical features with findings in the urine.
4. Most urinary pathogens can reduce nitrate to nitrite, and neutrophils and nitrites can usually be detected in symptomatic infections by urine dipstick tests for leucocyte esterase and nitrite, respectively.
5. The absence of both nitrites and leucocyte esterase in the urine makes UTI unlikely. Interpretation of bacterial counts in the urine, and of what is a 'significant' culture result, is based on probabilities.
6. Urine taken by suprapubic aspiration should be sterile, so the presence of any organisms is significant.
7. If the patient has symptoms and there are neutrophils in the urine, a small number of organisms is significant.
8. In asymptomatic patients $> 10^9$ organisms/ml is usually regarded as significant (asymptomatic bacteriuria).
9. Typical organisms causing UTI in the community include *E. coli* derived from the gastrointestinal tract (about 75% of infections), *Proteus spp.*, *Pseudomonas spp.*, streptococci and *Staphylococcus epidermidis*.
10. In hospital, *E. coli* still predominates, but *Klebsiella* or streptococci are more common.
11. Certain strains of *E. coli* have a particular propensity to invade the urinary tract.

Investigations to detect underlying predisposing factors for UTI are used selectively, most commonly in patients with recurrent infections.

Investigation of patients with urinary tract infection

All patients :

- Dipstick* estimation of nitrite, leucocyte esterase and glucose
- Microscopy/cytometry of urine for white blood cells, organisms

- Urine culture
Infants, children, and anyone with fever or complicated infection :
- Full blood count; urea, electrolytes, creatinine

Blood cultures

Pyelonephritis; males; children; women with recurrent infections :

- Renal tract ultrasound or CT
- Pelvic examination in women, rectal examination in men

Continuing haematuria or other suspicion of bladder lesion :

- Cystoscoping

Prostatomegaly

Benign Prostatic Hyperplasia (BPH)

- Benign prostate enlargement (BPE) is the medical term to describe an enlarged prostate.
- Benign prostatic hyperplasia (BPH), also known as benign prostatic hypertrophy, is a histologic diagnosis characterized by proliferation of the cellular elements of the prostate, leading to an enlarged prostate gland.
- Chronic bladder outlet obstruction (BOO) secondary to BPH may lead to urinary retention, impaired kidney function, recurrent urinary tract infections, gross hematuria, and bladder calculi.
- Benign prostate enlargement (BPE)

Signs and symptoms

- Constriction for the flow of urine due to enlarged prostate
- Nerves within the prostate and bladder may also play a role in causing the following common symptoms:

Urinary frequency

Urinary urgency

Nocturia- Needing to get up frequently at night to urinate

Hesitancy - Difficulty initiating the urinary stream; interrupted, weak stream

Incomplete bladder emptying - The feeling of persistent residual urine, regardless of the frequency of urination

Straining - The need strain or push (Valsalva maneuver) to initiate and maintain urination in order to more fully empty the bladder

Decreased force of stream - The subjective loss of force of the urinary stream over time

Dribbling - The loss of small amounts of urine due to a poor urinary stream as well as weak urinary stream

Diagnosis

Digital rectal examination : The digital rectal examination (DRE) is an integral part of the evaluation in men with presumed BPH. With the DRE, the examiner can assess prostate size and contour, evaluate for nodules, and detect areas suggestive of malignancy.

Laboratory studies

Urinalysis - Examine the urine using dipstick methods and/or via centrifuged sediment evaluation to assess for the presence of blood, leukocytes, bacteria, protein, or glucose

Urine culture - This may be useful to exclude infectious causes of irritative voiding and is usually performed if the initial urinalysis findings indicate an abnormality

Prostate-specific antigen (PSA) - Although BPH does not cause prostate cancer, men at risk for BPH are also at risk for this disease and should be screened accordingly (although screening for prostate cancer remains controversial)

Electrolytes, blood urea nitrogen (BUN), and creatinine - These evaluations are useful screening tools for chronic kidney disease in patients who have high postvoid residual (PVR) urine volumes; however, a routine serum creatinine measurement is not indicated in the initial evaluation of men with lower urinary tract symptoms (LUTS) secondary to BPH.

Ultrasonography

Ultrasonography (abdominal, renal, transrectal)

is useful for helping to determine bladder and prostate size and the degree of hydronephrosis (if any) in patients with urinary retention or signs of kidney insufficiency. Generally, it is not indicated for the initial evaluation of uncomplicated LUTS.

Endoscopy of the lower urinary tract

Cystoscopy may be indicated in patients scheduled for invasive treatment or in whom a foreign body or malignancy is suspected. In addition, endoscopy may be indicated in patients with a history of sexually transmitted disease (eg, gonococcal urethritis), prolonged catheterization, or trauma.

IPSS/AUA-SI

The severity of BPH can be determined with the International Prostate Symptom Score (IPSS)/American Urological Association Symptom Index (AUA-SI) plus a disease-specific quality of life (QOL) question.

Other tests

Flow rate - Useful in the initial assessment and to help determine the patient's response to treatment

PVR urine volume - Used to gauge the severity of bladder decompensation; it can be obtained invasively with a catheter or noninvasively with a transabdominal ultrasonic scanner

Pressure-flow studies - Findings may prove useful for evaluating for BOO

Urodynamic studies - To help distinguish poor bladder contraction ability (detrusor underactivity) from BOO

Cytologic examination of the urine - May be considered in patients with predominantly irritative voiding symptoms

Nephrotic Syndrome

Nephrotic syndrome describes the onset of heavy proteinuria (>3.0 g/24 h), hypertension, hypercholesterolemia, hypoalbuminemia, edema/anasarca, and microscopic hematuria; if only large amounts of proteinuria are present without clinical manifestations, the condition is sometimes called nephrotic-range proteinuria. The glomerular

filtration rate (GFR) in these patients may initially be normal or, rarely, higher than normal, but with persistent hyperfiltration and continued nephron loss, it typically declines over months to years. Patients with a basement membrane syndrome either have genetically abnormal basement membranes (Alport's syndrome) or an autoimmune response to basement membrane collagen IV (Goodpasture's syndrome) associated with microscopic hematuria, mild to heavy proteinuria, and hypertension with variable elevations in serum creatinine. Glomerular-vascular syndrome describes patients with vascular injury producing hematuria and moderate proteinuria. Affected individuals can have vasculitis, thrombotic microangiopathy, antiphospholipid syndrome, or, more commonly, a systemic disease such as atherosclerosis, cholesterol emboli, hypertension, sickle cell anemia, and autoimmunity. Infectious disease-associated syndrome is most important if one has an international perspective. Save for subacute bacterial endocarditis in the Western Hemisphere, malaria and schistosomiasis may be the most common causes of glomerulonephritis throughout the world, closely followed by HIV and chronic hepatitis B and C. These infectious diseases produce a variety of inflammatory reactions in glomerular capillaries, ranging from nephrotic syndrome to acute nephritic injury, and urinalyses that demonstrate a combination of hematuria and proteinuria.

Nephrotic syndrome is kidney disease with proteinuria, hypoalbuminemia, and edema. Nephrotic-range proteinuria is 3 grams per day or more. On a single spot urine collection, it is 2 g of protein per gram of urine creatinine.

Nephrotic syndrome may affect adults and children, of both sexes and of any race. It may occur in typical form, or in association with nephritic syndrome. The latter connotes glomerular inflammation, with hematuria and impaired kidney function.

Classification

Nephrotic syndrome can be primary; being a disease specific to the kidneys, or it can be

secondary, being a renal manifestation of a systemic general illness. In all cases, injury to glomeruli is an essential feature.

Primary causes of nephrotic syndrome include the following, in approximate order of frequency:

- Minimal-change nephropathy
- Focal glomerulosclerosis
- Membranous nephropathy
- Hereditary nephropathies

Secondary causes include the following, again in order of approximate frequency:

- Diabetes mellitus
- Lupus erythematosus
- Amyloidosis and paraproteinemias
- Viral infections (eg, hepatitis B, hepatitis C, human immunodeficiency virus [HIV])
- Preeclampsia

Nephrotic-range proteinuria may occur in other kidney diseases, such as IgA nephropathy. In that common glomerular disease, one third of patients may have nephrotic-range proteinuria.

Nephrotic syndrome may occur in persons with sickle cell disease and evolve to renal failure. Membranous nephropathy may complicate bone marrow transplantation, in association with graft versus host disease.

Kidney diseases that affect tubules and interstitium, such as interstitial nephritis, will not cause nephrotic syndrome.

Nephrotic syndrome classically presents with heavy proteinuria, minimal hematuria, hypoalbuminemia, hypercholesterolemia, edema, and hypertension. If left undiagnosed or untreated, some of these syndromes will progressively damage enough glomeruli to cause a fall in GFR, producing renal failure.

Clinical Features

The first sign of nephrotic syndrome in children is usually swelling of the face; this is followed by swelling of the entire body. Adults can present with dependent edema. Foamy urine may be a presenting feature. A thrombotic complication,

such as deep venous thrombosis of the calf veins or even a pulmonary embolus, may be the first clue indicating nephrotic syndrome.

Additional historical features can be related to the cause of nephrotic syndrome. Thus, the recent start of a nonsteroidal anti-inflammatory drug (NSAID) or a 10-year history of diabetes may be very relevant.

Difference between acute nephritis and nephrotic syndrome

Acute nephritis	Nephrotic syndrome
• Characterized by hematuria, edema, hypertension, oliguria	• Characterized by heavy proteinuria, hpo albuminemia, edema, hperlipidemia
• 90% post infective immune complex	• 90% idiopathic
• Usually only 1 attack	• Relapses common
• Immune complex deposition	• Retraction of epithelial foot process
• Urine: Alb 1+/2+, hematuria, RBC cast	• Urine: Selective proteinuria, No RBC
• Blood urea/creat raised	• Blood urea/ creat normal

Differentiation Between Nephrotic Syndrome and Nephritic Syndrome

Onset	Insidious	Abrupt
Edema	++	+++++
Blood pressure	Normal	Raised
Jugular venous pressure	Normal/low	Raised
Proteinuria	++	+++++
Nematuria	May/may not occur	+++
Red blood cell casts	Absent	Present
Serum albumin	Low	Normal/slightly reduced

Diagnostic Criteria-Nephritic Syndrome

- Inflammatory rupture of glomerular capillaries resulting in bleeding into urinary space

(Bowman's capsule). Proteinuria and oedema mild.

- Oliguria - Reduced GFR causing reduced urine output.
- Azotemia - elevated BUN.
- Hypertension - increased fluid retention & Renin-Angiotension-Aldosterone activation by ischaemic kidneys
- Hematuria - form RBC casts & granular casts

Diagnostic Criteria for Nephrotic Syndrome

Factor	Criteria
Heavy proteinuria	Spot urine showing a protein-to-creatinine ratio of > 3 to 3.5 mg protein/mg creatinine (300 to 350 mg/mmol), or 24-hour urine collection showing > 3 to 3.5 g protein
Hypoalbuminemia	Serum albumin < 2.5 g per dL (25 g per L)
Edema	Clinical evidence of peripheral edema
Hyperlipidemia (not required)	Severe hyperlipidemia, total cholesterol is often > 350 mg per DL (9.06 mmol per L)

Physical Examination

Edema is the predominant feature of nephrotic syndrome and initially develops around the eyes and legs. With time, the edema becomes generalized and may be associated with an increase in weight, the development of ascites, or pleural effusions. Hematuria and hypertension manifest in a minority of patients.

Additional features on examination will vary according to cause and as a result of whether or not renal function impairment exists. Thus, in the case of longstanding diabetes, there may be diabetic retinopathy, which correlates closely with diabetic nephropathy. If the kidney function is reduced, there may be hypertension and/or anemia.

Ancillary Investigations

Diagnostic studies for nephrotic syndrome may include the following:

- Urinalysis

- Urine sediment examination
- Urinary protein measurement
- Serum albumin
- Serologic studies for infection and immune abnormalities
- Renal ultrasonography
- Renal biopsy

Urine Studies

Urinalysis: Urinalysis is the first test used in the diagnosis of nephrotic syndrome. Nephrotic range proteinuria will be apparent by 3+ or 4+ readings on the dipstick, or by semiquantitative testing by sulfosalicylic acid. A 3+ reading represents 300 mg/dL of urinary protein or more, which is 3 g/L or more and thus in the nephrotic range.

Nephritis

Nephritis is a condition in which the nephrons, the functional units of the kidneys, become inflamed. This inflammation, which is also known as glomerulonephritis, can adversely affect kidney function.

Causes

Nephritis is often caused by infections, and toxins. In some cases, the cause may not be clear.

Nephritis and kidney disease often seem to run in families, which suggests a possible genetic component. Some infections, such as HIV and hepatitis B or C, can also cause nephritis.

In some cases, kidney damage can occur as a result of medications, such as antibiotics. This damage can lead to nephritis. Taking too many pain relievers, nonsteroidal anti-inflammatory drugs (NSAIDs), or diuretic pills can also cause this condition. Pyelonephritis is inflammation that results from a urinary tract infection that reaches the renal pelvis of the kidney.

Lupus nephritis is inflammation of the kidney caused by systemic lupus erythematosus (SLE), a disease of the immune system.

Athletic nephritis is nephritis resulting from strenuous exercise. Bloody urine after strenuous

exercise may also result from March hemoglobinuria, which is caused by trauma to red blood cells, causing their rupture, which leads to the release of hemoglobin into the urine.

Risk factors : The most important risk factors for kidney disease are:

- A family history of kidney disease
- High blood pressure
- Diabetes
- Obesity
- Heart disease
- An age of 60 years or more

Types of Nephritis

There are several different types of nephritis, including:

Acute glomerulonephritis: This form of nephritis can develop suddenly after a severe infection, such as strep throat, hepatitis, or HIV. Lupus and rarer disorders, such as vasculitides and granulomatosis with polyangiitis (GPA), can also lead to acute inflammation of the kidneys. A person with these conditions will require prompt medical attention during a flare-up to reduce kidney damage.

Lupus nephritis: Lupus is an autoimmune disease, which means that the immune system mistakenly attacks healthy tissues in the body. Over half of all individuals with a lupus diagnosis eventually develop lupus nephritis. This occurs when the immune system attacks the kidneys.

Alport syndrome, or hereditary nephritis: This disease can lead to kidney failure, as well as vision and hearing problems. Alport syndrome is passed on in the genes, and it is usually more severe in men.

Chronic glomerulonephritis: This form of nephritis develops slowly and causes few symptoms in its early stages. As with acute glomerulonephritis, this condition can cause severe kidney damage and kidney failure. It may run in families or develop after a sudden disease.

IgA nephropathy: This is one of the more common forms of nephritis. It develops when IgA

antibody deposits build up in the kidneys and cause inflammation. The immune system develops antibodies to combat harmful substances and organisms that enter the body. People with IgA neuropathy have defective IgA antibodies.

Interstitial nephritis: Often developing very rapidly, this form of nephritis usually occurs due to infection or a particular medication. It affects the part of the kidney called the interstitium, which is a fluid-filled space.

General symptoms

The symptoms of nephritis are rarely severe in the early stages. The following signs may indicate that a person has this condition:

- Changes in urinating habits
- Swelling anywhere in the body, especially the hands, feet, ankles, and face
- Changes in urine color
- Foamy urine
- Blood in the urine

Lupus nephritis

Lupus nephritis is histologically evident in most patients with systemic lupus erythematosus (SLE), even those without clinical manifestations of renal disease. Evaluating renal function in SLE patients is important because early detection and treatment of renal involvement can significantly improve renal outcome.

Aetiology

Genetic factors: As with many autoimmune disorders, evidence suggests that genetic predisposition plays an important role in the development of both SLE and lupus nephritis. Multiple genes, many of which are not yet identified, mediate this genetic predisposition.

Signs and symptoms

Patients with lupus nephritis may report the following :

- Other symptoms of active SLE (eg, fatigue, fever, rash, arthritis, serositis, or central nervous system [CNS] disease); these are more

common with focal proliferative and diffuse proliferative lupus nephritis

- Asymptomatic lupus nephritis - During regular follow-up, laboratory abnormalities suggest active lupus nephritis; this is more typical of mesangial or membranous lupus nephritis.
- Active nephritis - Peripheral edema secondary to hypertension or hypoalbuminemia; extreme peripheral edema is more common with diffuse or membranous lupus nephritis.
- Diffuse lupus nephritis - Headache, dizziness, visual disturbances, or signs of cardiac decompensation

Physical findings may include the following:

- Focal and diffuse lupus nephritis - Generalized active SLE with the presence of a rash, oral or nasal ulcers, synovitis, or serositis; signs of active nephritis
- Active lupus nephritis - Hypertension, peripheral edema, and, occasionally, cardiac decompensation.
- Membranous lupus nephritis - Peripheral edema, ascites, and pleural and pericardial effusions without hypertension

Required Investigations

Laboratory tests to evaluate renal function in SLE patients include the following :

- Blood urea nitrogen (BUN) testing
- Serum creatinine assessment
- Urinalysis (to check for protein, red blood cells [RBCs], and cellular casts)
- Spot urine test for creatinine and protein concentration
- 24-hour urine test for creatinine clearance and protein excretion

Laboratory tests for SLE disease activity include the following:

- Antibodies to double-stranded DNA (dsDNA)
- Complement (C3, C4, and CH50)
- Erythrocyte sedimentation rate (ESR)
- C-reactive protein (CRP)

Renal biopsy should be considered in any

patient with SLE who has clinical or laboratory evidence of active nephritis, especially upon the first episode of nephritis.

Lupus nephritis is staged according to the classification revised by the International Society of Nephrology (ISN) and the Renal Pathology Society (RPS) in 2003, as follows :

- Class I - Minimal mesangial lupus nephritis
- Class II - Mesangial proliferative lupus nephritis
- Class III - Focal lupus nephritis (active and chronic; proliferative and sclerosing)
- Class IV - Diffuse lupus nephritis (active and chronic; proliferative and sclerosing; segmental and global)
- Class V - Membranous lupus nephritis
- Class VI - Advanced sclerosis lupus nephritis

Tubulointerstitial Nephritis

The term tubulointerstitial is used to broadly refer to kidney diseases that involve structures in the kidney outside the glomerulus. These diseases generally involve tubules and/or the interstitium of the kidney and spare the glomeruli.

Kidney biopsy : This is an example of acute interstitial nephritis. The renal cortex shows a diffuse interstitial, predominantly mononuclear, inflammatory infiltrate with no changes to the glomerulus. Tubules in the center of the field are separated by inflammation and edema, as compared with the more normal architecture in the right lower area (periodic acid-Schiff, 40 X).

Although primary glomerular diseases are often associated with prominent tubulointerstitial changes, the clinical presentation is dominated by the consequences of glomerular injury.

Kidney biopsy in interstitial nephritis. Acute crescentic glomerulonephritis. The glomerular tuft is compressed by the proliferation of epithelial cells, forming a crescent. The interstitium shows mononuclear inflammatory infiltrate and edema (periodic acid-Schiff, 40 X).

Aetiology

Tubulointerstitial diseases of the kidney

encompass diverse etiologies and pathophysiologic processes, and the patient can present with acute or chronic conditions. Many forms of tubulointerstitial injury involve exposure to drugs or other nephrotoxic agents such as heavy metals and, rarely, infection. By far the most common form of tubulointerstitial inflammation is hypersensitivity reaction to medications, termed allergic interstitial nephritis.

History and Physical Examination

The history of patients with tubulointerstitial nephritis depends on whether the disease is acute or chronic. A thorough physical examination may provide clues to the diagnosis (eg, fever, rash in acute tubulointerstitial nephritis, lividoreticularis and Hollenhorst plaques in the optic fundi in atheroembolic disease), but, in most patients, no characteristic findings exist. Some patients present with hypertension, although others may be normotensive or hypotensive (eg, Balkan endemic nephropathy).

Investigations

CBC With Differential: Eosinophilia, when present, can be very helpful in the evaluation of tubulointerstitial nephritis. Typically, eosinophilia is absent in acute tubulointerstitial nephritis that is induced by nonsteroidal anti-inflammatory drugs (NSAIDs).

Chemistry Panel

A complete set of chemistries, including blood urea nitrogen (BUN) and serum creatinine, provides information on whether renal insufficiency exists. A low bicarbonate level (total carbon dioxide < 24-23 mEq/L) may indicate acidosis. Low serum potassium levels may indicate a proximal tubular disorder, and elevated serum potassium levels with a low bicarbonate level may indicate type 4 renal tubular acidosis, which can be observed with lead nephropathy and nonsteroidal anti-inflammatory drug (NSAID)-induced analgesic nephropathy, among other conditions.

Urine Studies

Urinalysis may reveal proteinuria, hematuria,

and the presence of white blood cells (WBCs), with or without bacteria. A microscopic analysis of urine sediment may reveal casts, WBCs, eosinophils, and crystals. If allergic interstitial nephritis is suspected, send a cytospin specimen to determine if eosinophils are in the urine. Unfortunately, the absence of eosinophiluria does not rule out the diagnosis, and it can be observed in other diseases, including cholesterol microembolism, urinary tract infections, parasitic disorders, and glomerulonephritis. In nonsteroidal anti-inflammatory drug (NSAID)-induced acute tubulointerstitial nephritis, eosinophiluria is usually absent.

Ultrasonography and Radiography

Ultrasonography is noninvasive imaging technique that is extremely helpful in identifying hydronephrosis in obstructive disease as well as calculi in stone disease. Both radiolucent and radiopaque stones can be visualized with this modality. A combination of ultrasonography and flat plate kidney, ureter, and bladder (KUB) radiography is helpful in the workup and identification of radiopaque versus radiolucent stones.

Kidney Biopsy and Histologic Features

Kidney biopsy is the definitive test for diagnosing acute allergic interstitial nephritis, particularly in cases in which the clinical diagnosis is difficult. Because the differential diagnosis of acute tubulointerstitial nephritis encompasses multiple etiologies, consider kidney biopsy when the diagnosis is not obvious.

Kidney biopsy shows mononuclear and often eosinophilic cellular infiltration of the renal parenchyma with sparing of the glomeruli. Sometimes, interstitial changes such as fibrosis and atrophy are also present (eg, Renal biopsies have shown severe interstitial fibrosis in patients with Chinese herb/aristolochic acid nephropathy).

Acute Kidney Injury

Acute kidney injury (AKI), previously known as acute renal failure, is characterized by the sudden impairment of kidney function resulting in the retention of nitrogenous and other waste products

normally cleared by the kidneys. AKI is not a single disease but, rather, a designation for a heterogeneous group of conditions that share common diagnostic features: specifically, an increase in the blood urea nitrogen (BUN) concentration and/or an increase in the plasma or serum creatinine (SCr) concentration often associated with a reduction in urine volume. AKI can range in severity from asymptomatic and transient changes in laboratory parameters of glomerular filtration rate (GFR), to overwhelming and rapidly fatal derangements in effective circulating volume regulation and electrolyte and acid-base composition of the plasma.

Changing the name of a syndrome as well known as "acute renal failure" does not occur frequently. We will summarize some of the reasons why the name was changed to "acute kidney injury." The term failure reflects only part of the spectrum of damage to the kidney that occurs clinically. In most cases of damage, the reduction in kidney function is modest. Nevertheless, this modest change has been documented to be associated with negative effects on outcome, albeit not nearly as ominous as seen with large decreases in kidney function associated with frank kidney failure that often requires acute dialysis therapies. Furthermore, the term renal is not well understood in the general population and this makes communication with patients and family more challenging; hence "kidney" has replaced "renal."

Categories of AKI

AKI may be classified into 3 general categories, as follows:

- Prerenal - As an adaptive response to severe volume depletion and hypotension, with structurally intact nephrons
- Intrinsic - In response to cytotoxic, ischemic, or inflammatory insults to the kidney, with structural and functional damage
- Postrenal - From obstruction to the passage of urine

Etiology

The etiology of AKI consists of 3 main mechanisms: prerenal, intrinsic, and obstructive.

Prerenal AKI

Prerenal AKI represents the most common form of kidney injury and often leads to intrinsic AKI if it is not promptly corrected. Volume loss can provoke this syndrome.

Volume depletion can be caused by the following:

- Renal losses - Diuretics, polyuria
- GI losses - Vomiting, diarrhea
- Cutaneous losses - Burns, Stevens-Johnson syndrome
- Hemorrhage
- Pancreatitis

Decreased cardiac output can be caused by the following:

- Heart failure
- Pulmonary embolus
- Acute myocardial infarction
- Severe valvular disease
- Abdominal compartment syndrome - Tense ascites

Systemic vasodilation can be caused by the following:

- Sepsis
- Anaphylaxis
- Anesthetics
- Drug overdose

Afferent arteriolar vasoconstriction can be caused by the following:

- Hypercalcemia
- Drugs - NSAIDs, amphotericin B, calcineurin inhibitors, norepinephrine, radiocontrast agents
- Hepatorenal syndrome

Diseases that decrease effective arterial blood volume include the following:

- Hypovolemia
- Heart failure
- Liver failure
- Sepsis

Renal arterial diseases that can result in AKI include renal arterial stenosis, especially in the

setting of hypotension or initiation of ACE inhibitors or ARBs. Renal artery stenosis typically results from atherosclerosis or fibromuscular dysplasia, but is also a feature of the genetic syndromes type 1 neurofibromatosis, Williams syndrome, and Alagille syndrome.

Patients can also develop septic embolic disease (eg, from endocarditis) or cholesterol emboli, often as a result of instrumentation or cardiovascular surgery.

Intrinsic AKI

Structural injury in the kidney is the hallmark of intrinsic AKI.

Vascular (large- and small-vessel) causes of intrinsic AKI include the following :

- Renal artery obstruction - Thrombosis, emboli, dissection, vasculitis
- Renal vein obstruction - Thrombosis
- Microangiopathy - Thrombotic thrombocytopenic purpura (TTP), Hemolytic-uremic syndrome (HUS), disseminated intravascular coagulation (DIC), preeclampsia
- Malignant hypertension
- Scleroderma renal crisis
- Transplant rejection
- Atheroembolic disease

Glomerular causes include the following:

- Anti-glomerular basement membrane (GBM) disease - As part of Goodpasture syndrome or renal limited disease
- Anti-neutrophil cytoplasmic antibody-associated glomerulonephritis (ANCA-associated glomerulonephritis) - Wegener granulomatosis, Churg-Strauss syndrome, microscopic polyangiitis
- Immune complex glomerulonephritis - Lupus, postinfectious glomerulonephritis, cryoglobulinemia, primary membranoproliferative glomerulonephritis

Tubular etiologies may include ischemia or cytotoxicity. Cytotoxic etiologies include the following:

- Heme pigment - Rhabdomyolysis, intravascular hemolysis
- Crystals - Tumor lysis syndrome, seizures, ethylene glycol poisoning, megadose vitamin C, acyclovir, indinavir, methotrexate
- Drugs - Aminoglycosides, lithium, amphotericin B, pentamidine, cisplatin, ifosfamide, radiocontrast agents

Interstitial causes include the following:

- Drugs - Penicillins, cephalosporins, NSAIDs, proton-pump inhibitors, allopurinol, rifampin, indinavir, mesalamine, sulfonamides
- Infection - Pyelonephritis, viral nephritides
- Systemic disease - Sjögren syndrome, sarcoid, lupus, lymphoma, leukemia, tubulonephritis, uveitis

Postrenal AKI

Mechanical obstruction of the urinary collecting system, including the renal pelvis, ureters, bladder, or urethra, results in obstructive uropathy or postrenal AKI.

Causes of obstruction include the following:

Causes of postrenal AKI include the following:

- Ureteric obstruction - Stone disease, tumor, fibrosis, ligation during pelvic surgery
- Bladder neck obstruction - Benign prostatic hypertrophy (BPH), cancer of the prostate (CA prostate or prostatic CA), neurogenic bladder, tricyclic antidepressants, ganglion blockers, bladder tumor, stone disease, hemorrhage/clot
- Urethral obstruction - Strictures, tumor, phimosis
- Intra-abdominal hypertension - Tense ascites
- Renal vein thrombosis

Diseases causing urinary obstruction from the level of the renal tubules to the urethra include the following:

- Tubular obstruction from crystals - Eg, uric acid, calcium oxalate, acyclovir, sulfonamide, methotrexate, myeloma light chains

- Ureteral obstruction - Retroperitoneal tumor, retroperitoneal fibrosis (methysergide, propranolol, hydralazine), urolithiasis, or papillary necrosis
- Urethral obstruction - Benign prostatic hypertrophy; prostate, cervical, bladder, or colorectal carcinoma; bladder hematoma; bladder stone; obstructed Foley catheter; neurogenic bladder; stricture

Etiology in newborns and infants

Prerenal AKI

The patient's age has significant implications for the differential diagnosis of AKI. In newborns and infants, causes of prerenal AKI include the following:

- Perinatal hemorrhage - Twin-twin transfusion, complications of amniocentesis, abruptio placenta, birth trauma
- Neonatal hemorrhage - Severe intraventricular hemorrhage, adrenal hemorrhage
- Perinatal asphyxia and hyaline membrane disease (newborn respiratory distress syndrome) - Both may result in preferential blood shunting away from the kidneys (ie, prerenal) to central circulation

Intrinsic AKI

Causes of intrinsic AKI include the following:

- ATN - Can occur in the setting of perinatal asphyxia; ATN also has been observed secondary to medications (eg, aminoglycosides, NSAIDs) given to the mother perinatally
- ACE inhibitors - Can traverse the placenta, resulting in a hemodynamically mediated form of AKI
- Acute glomerulonephritis - Rare; most commonly the result of maternal-fetal transfer of antibodies against the neonate's glomeruli or transfer of chronic infections (syphilis, cytomegalovirus) associated with acute glomerulonephritis

Postrenal AKI

Congenital malformations of the urinary

collecting systems should be suspected in cases of postrenal AKI.

Pathophysiology

The driving force for glomerular filtration is the pressure gradient from the glomerulus to the Bowman space. Glomerular pressure depends primarily on renal blood flow (RBF) and is controlled by the combined resistances of renal afferent and efferent arterioles. Regardless of the cause of AKI, reductions in RBF represent a common pathologic pathway for decreasing glomerular filtration rate (GFR).

Clinical Features

Prerenal azotemia should be suspected in the setting of vomiting, diarrhea, glycosuria causing polyuria, and several medications including diuretics, NSAIDs, ACE inhibitors, and ARBs.

Physical Examination

Physical signs of orthostatic hypotension, tachycardia, reduced jugular venous pressure, decreased skin turgor, and dry mucous membranes are often present in prerenal azotemia.

A history of prostatic disease, nephrolithiasis, or pelvic or paraaortic malignancy would suggest the possibility of postrenal AKI. Whether or not symptoms are present early during obstruction of the urinary tract depends on the location of obstruction. Colicky flank pain radiating to the groin suggests acute ureteric obstruction. Nocturia and urinary frequency or hesitancy can be seen in prostatic disease. Abdominal fullness and suprapubic pain can accompany massive bladder enlargement. Definitive diagnosis of obstruction requires radiologic investigations.

A careful review of all medications is imperative in the evaluation of an individual with AKI. Not only are medications frequently a cause of AKI but doses of administered medications must be adjusted for estimated GFR. Idiosyncratic reactions to a wide variety of medications can lead to allergic interstitial nephritis, which may be accompanied by fever, arthralgias, and a pruritic erythematous rash. The absence of systemic features of

hypersensitivity, however, does not exclude the diagnosis of interstitial nephritis.

AKI accompanied by palpable purpura, pulmonary hemorrhage, or sinusitis raises the possibility of systemic vasculitis with glomerulonephritis. Atheroembolic disease can be associated with livedo reticularis and other signs of emboli to the legs. A tense abdomen should prompt consideration of acute abdominal compartment syndrome, which requires measurement of bladder pressure. Signs of limb ischemia may be clues to the diagnosis of rhabdomyolysis.

Urine Findings

Complete anuria early in the course of AKI is uncommon except in the following situations: complete urinary tract obstruction, renal artery occlusion, overwhelming septic shock, severe ischemia (often with cortical necrosis), or severe proliferative glomerulonephritis or vasculitis.

A reduction in urine output (oliguria, defined as <400 mL/24 h) usually denotes more significant AKI (i.e., lower GFR) than when urine output is preserved.

Oliguria is associated with worse clinical outcomes. Preserved urine output can be seen in nephrogenic diabetes insipidus characteristic of long standing urinary tract obstruction, tubulointerstitial disease, or nephrotoxicity from cisplatin or aminoglycosides, among other causes. Red or brown urine may be seen with or without gross hematuria; if the color persists in the supernatant after centrifugation, then pigment nephropathy from rhabdomyolysis or hemolysis should be suspected.

The urinalysis and urine sediment examination are invaluable tools, but they require clinical correlation because of generally limited sensitivity and specificity. In the absence of preexisting proteinuria from CKD, AKI from ischemia or nephrotoxins leads to mild proteinuria (<1 g/d). Greater proteinuria in AKI suggests damage to the glomerular ultrafiltration barrier or excretion of myeloma light chains; the latter are not detected

with conventional urine dipsticks (which detect albumin) and require the sulfosalicylic acid test or immunoelectrophoresis. Atheroemboli can cause a variable degree of proteinuria. Extremely heavy proteinuria ("nephrotic range," >3.5 g/d) can occasionally be seen in glomerulonephritis, vasculitis, or interstitial nephritis (particularly from NSAIDs). AKI can also complicate cases of minimal change disease, a cause of the nephrotic syndrome. If the dipstick is positive for hemoglobin but few red blood cells are evident in the urine sediment, then rhabdomyolysis or hemolysis should be suspected.

Physical Examination

Obtaining a thorough physical examination is extremely important when collecting evidence about the etiology of AKI. Clues may be found in any of the following: Skin; Eyes; Ears; Cardiovascular system; Abdomen and Pulmonary system.

Skin

Skin examination may reveal the following:

- Livedo reticularis, digital ischemia, butterfly rash, palpable purpura - Systemic vasculitis
- Maculopapular rash - Allergic interstitial nephritis
- Track marks (ie, intravenous drug abuse) - Endocarditis

Petechiae, purpura, ecchymosis, and livedo reticularis provide clues to inflammatory and vascular causes of AK. Infectious diseases, thrombotic thrombocytopenic purpura (TTP), disseminated intravascular coagulation (DIC), and embolic phenomena can produce typical cutaneous changes.

Eyes and ears

Eye examination may reveal the following:

- Keratitis, iritis, uveitis, dry conjunctivae - Autoimmune vasculitis
- Jaundice - Liver diseases
- Band keratopathy (ie, hypercalcemia) - Multiple myeloma

- Signs of diabetes mellitus
- Signs of hypertension
- Atheroemboli - Retinopathy

Evidence of uveitis may indicate interstitial nephritis and necrotizing vasculitis. Ocular palsy may indicate ethylene glycol poisoning or necrotizing vasculitis. Findings suggestive of severe hypertension, atheroembolic disease, and endocarditis may be observed on careful examination of the eyes.

Ear examination may reveal the following:

- Hearing loss - Alport disease and aminoglycoside toxicity
- Mucosal or cartilaginous ulcerations - Wegener granulomatosis

Cardiovascular system

The most important part of the physical examination is the assessment of cardiovascular and volume status. The physical examination must include the following:

- Pulse rate and blood pressure recordings measured in the supine and the standing position
- Close inspection of the jugulovenous pulse
- Careful examination of the heart and lungs, skin turgor, and mucous membranes
- Assessment for peripheral edema

Cardiovascular examination may reveal the following:

- Irregular rhythms (ie, atrial fibrillation) - Thromboemboli
- Murmurs - Endocarditis
- Pericardial friction rub - Uremic pericarditis
- Increased jugulovenous distention, rales, S_3 - Heart failure

In hospitalized patients, accurate daily records of fluid intake and urine output, as well as daily measurements of patient weight, are important. Hypovolemia leads to hypotension; however, hypotension may not necessarily indicate hypovolemia.

Severe heart failure may also cause hypotension. Although patients with heart failure

may have low blood pressure, volume expansion is present and effective renal perfusion is poor, which can result in AKI.

Severe hypertension with renal failure suggests one of the following disorders:

- Renovascular disease
- Glomerulonephritis
- Vasculitis
- Atheroembolic disease

Abdomen

Abdominal examination may reveal the following:

- Pulsatile mass or bruit - Atheroemboli
- Abdominal or costovertebral angle tenderness - Nephrolithiasis, papillary necrosis, renal artery thrombosis, renal vein thrombosis
- Pelvic, rectal masses; prostatic hypertrophy; distended bladder - Urinary obstruction
- Limb ischemia, edema - Rhabdomyolysis

Abdominal examination findings can be useful in helping to detect obstruction at the bladder outlet as the cause of renal failure; such obstruction may be due to cancer or to an enlarged prostate.

The presence of tense ascites can indicate elevated intra-abdominal pressure that can retard renal venous return and result in AKI. The presence of an epigastric bruit suggests renal vascular hypertension, which may predispose to AKI.

Pulmonary system

Pulmonary examination may reveal the following:

- Rales - Goodpasture syndrome, Wegener granulomatosis
- Hemoptysis - Wegener granulomatosis

Ancillary Investigations

Several laboratory tests, including the following, are useful for assessing the etiology of acute kidney injury (AKI) and can aid in proper management of the disease:

- Complete blood count (CBC)
- Serum biochemistries

- Urine analysis with microscopy
- Urine electrolytes

In some cases, renal imaging is useful, especially if renal failure is secondary to obstruction. The American College of Radiology recommends ultrasonography, preferably with Doppler methods, as the most appropriate imaging method in AKI.

Kidney Function Studies

Although increased levels of blood urea nitrogen (BUN) and creatinine are the hallmarks of renal failure, the rate of rise depends on the degree of renal insult and, with respect to BUN, on protein intake. BUN may be elevated in patients with gastrointestinal (GI) or mucosal bleeding, steroid treatment, or protein loading.

The ratio of BUN to creatinine is an important finding. The ratio can exceed 20:1 in conditions in which enhanced reabsorption of urea is favored (eg, in volume contraction); this suggests prerenal AKI.

Urinalysis

Findings of granular, muddy brown casts are highly suggestive of tubular necrosis. The presence of tubular cells or tubular cell casts also supports the diagnosis of ATN. Often, oxalate crystals are observed in cases of ATN.

Fractional Excretion of Sodium and Urea

Sodium : Urine electrolyte findings also can serve as valuable indicators of functioning renal tubules. The fractional excretion of sodium (FENa) is the commonly used indicator. However, the interpretation of results from patients in nonoliguric states, those with glomerulonephritis, and those receiving or ingesting diuretics can lead to an erroneous diagnosis.

FENa can be a valuable test for helping to detect extreme renal avidity for sodium in conditions such as hepatorenal syndrome.

Urea

In patients who are receiving diuretics, a fractional excretion of urea (FEUrea) can be

obtained, since urea transport is not affected by diuretics.

Bladder Pressure

An intra-abdominal pressure of less than 10 mm Hg is considered normal and suggests that abdominal compartment syndrome is not the cause of AKI. An intra-abdominal pressure above 10 mm Hg is abnormal, but patients who have pressures of 15-25 mm Hg are at particular risk for abdominal compartment syndrome, and those with bladder pressures above 25 mm Hg should be suspected of having AKI as a result of abdominal compartment syndrome.

Emerging Biomarkers

Creatinine elevation is a late marker for renal dysfunction and, once elevated, reflects a severe reduction in glomerular filtration rate (GFR). Consequently, a number of biomarkers are being investigated to risk stratify and predict AKI in patients at risk for the disease.

Ultrasonography

Renal ultrasonography is useful for evaluating existing renal disease and obstruction of the urinary collecting system. Obtaining images of the kidneys can be technically difficult in patients who are obese, however, as well as in those with abdominal distention from ascites, gas, or retroperitoneal fluid collection.

Doppler ultrasonography

Doppler scans are useful for detecting the presence and nature of renal blood flow. Because renal blood flow is reduced in prerenal and intrarenal AKI, findings are of little use in the diagnosis of AKI.

Nuclear Scanning

Radionuclide imaging with technetium-99m-mercaptoacetyl triglycine (^{99m}Tc -MAG3), ^{99m}Tc -diethylene-triamine penta-acetic acid (^{99m}Tc -DTPA), or iodine-131 (^{131}I)-hippurate can be used to assess renal blood flow, as well as tubular function.

Aortorenal Angiography

Aortorenal angiography can be helpful in

establishing the diagnosis of renal vascular diseases, including the following:

- Renal artery stenosis
- Renal atheroembolic disease
- Atherosclerosis with aortorenal occlusion
- Certain cases of necrotizing vasculitis (eg, polyarteritis nodosa)

Renal Biopsy

A renal biopsy may also be indicated when renal function does not return for a prolonged period and a prognosis is required to develop long-term management. In as many as 40% of cases, renal biopsy results reveal an unexpected diagnosis.

Chronic Kidney Disease

Chronic kidney disease (CKD) encompasses a spectrum of different pathophysiologic processes associated with abnormal kidney function and a progressive decline in glomerular filtration rate (GFR). The term chronic renal failure applies to the process of continuing significant irreversible reduction in nephron number and typically corresponds to CKD stages 3–5. The pathophysiologic processes and adaptations associated with chronic renal failure. The dispiriting term end-stage renal disease represents a stage of CKD where the accumulation of toxins, fluid, and electrolytes normally excreted by the kidneys results in the uremic syndrome. This syndrome leads to death unless the toxins are removed by renal replacement therapy, using dialysis or kidney transplantation.

Pathophysiology of Chronic Kidney Disease

The pathophysiology of CKD involves two broad sets of mechanisms of damage: (1) initiating mechanisms specific to the underlying etiology (e.g., genetically determined abnormalities in kidney development or integrity, immune complex deposition and inflammation in certain types of glomerulonephritis, or toxin exposure in certain diseases of the renal tubules and interstitium) and (2) a set of progressive mechanisms, involving hyperfiltration and hypertrophy of the remaining

viable nephrons, that are a common consequence following long-term reduction of renal mass, irrespective of underlying etiology. The responses to reduction in nephron number are mediated by vasoactive hormones, cytokines, and growth factors. Eventually, these short-term adaptations of hypertrophy and hyperfiltration become maladaptive as the increased pressure and flow predisposes to distortion of glomerular architecture, associated with sclerosis and dropout of the remaining nephrons. Increased intrarenal activity of the renin-angiotensin axis appears to contribute both to the initial adaptive hyperfiltration and to the subsequent maladaptive hypertrophy and sclerosis, the latter, in part, owing to the stimulation of transforming growth factor (TGF- α).

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of the renin-angiotensin axis appears to contribute both to the initial adaptive hyperfiltration and to the subsequent maladaptive hypertrophy and sclerosis, the latter, in part, owing to the stimulation of transforming growth factor (TGF- β). This process explains why a reduction in renal mass from an isolated insult may lead to a progressive decline in renal function over many years.

Identification of Risk Factors and Staging of CKD.

It is important to identify factors that increase the risk for CKD, even in individuals with normal GFR. Risk factors include hypertension, diabetes mellitus, autoimmune disease, older age, African ancestry, a family history of renal disease, a previous episode of acute kidney injury, and the presence of proteinuria, abnormal urinary sediment, or structural abnormalities of the urinary tract. Recent research in the genetics of predisposition to common complex diseases has revealed DNA sequence variants at a number of genetic loci that are associated with common forms of CKD. In order to stage CKD, it is necessary to estimate the GFR.

Pathophysiology and Biochemistry of Uremia

Although serum urea and creatinine concentrations are used to measure the excretory capacity of the kidneys, accumulation of these two molecules themselves do not account for the many symptoms and signs that characterize the uremic syndrome in advanced renal failure. Hundreds of toxins that accumulate in renal failure have been implicated in the uremic syndrome. These include water-soluble, hydrophobic, protein-bound, charged, and uncharged compounds. Additional categories of nitrogenous excretory products include guanidino compounds, urates and hippurates, products of nucleic acid metabolism, polyamines, myoinositol, phenols, benzoates, and indoles. Compounds with a molecular mass between 500 and 1500 Da, the so-called middle molecules, are also retained and contribute to morbidity and mortality. It is thus evident that the serum concentrations of urea and creatinine should

be viewed as being readily measured, but incomplete, surrogate markers for these compounds, and monitoring the levels of urea and creatinine in the patient with impaired kidney function represents a vast oversimplification of the uremic state.

Clinical Features Chronic Kidney Disease And Uremia

Uremia leads to disturbances in the function of virtually every organ system. Chronic dialysis can reduce the incidence and severity of many of these disturbances, so that the overt and florid manifestations of uremia have largely disappeared in the modern health setting.

Fluid and electrolyte disturbances: Volume expansion, Hyponatremia, Hyperkalemia and Hyperphosphatemia.

Endocrine-metabolic disturbances: Secondary hyperparathyroidism, Adynamic bone, Vitamin D-deficient, osteomalacia, Carbohydrate resistance, Hyperuricemia, Hypertriglyceridemia, Increased Lp(a) level, Decreased high-density lipoprotein level, Protein-energy malnutrition, Impaired growth and development, Infertility and sexual dysfunction, and Amenorrhea.

Neuromuscular disturbances: Fatigue, Sleep disorders, Headache, Impaired mentation, Lethargy, Asterixis, Muscular irritability, Peripheral neuropathy, Restless legs syndrome, Myoclonus, Seizures, Coma, Muscle cramps, Dialysis disequilibrium syndrome and Myopathy.

Cardiovascular and pulmonary disturbances: Arterial hypertension, Congestive heart failure or pulmonary edema, Pericarditis, Hypertrophic or dilated cardiomyopathy, Uremic lung, Accelerated atherosclerosis, Hypotension and arrhythmias and vascular calcification.

Dermatologic disturbances: Pallor, Hyperpigmentation, Pruritus, Ecchymoses, Nephrogenic fibrosis, dermatopathy and Uremic frost.

Gastrointestinal disturbances: Anorexia, Nausea and vomiting, Gastroenteritis, Peptic ulcer, Gastrointestinal bleeding, Idiopathic ascites and Peritonitis.

Hematologic and immunologic disturbances: Anemia, Lymphocytopenia, Bleeding diathesis, increased susceptibility to infection, Leukopenia and Thrombocytopenia.

History and Physical Examination

Symptoms and overt signs of kidney disease are often subtle or absent until renal failure supervenes. Thus, the diagnosis of kidney disease often surprises patients and may be a cause of skepticism and denial. Particular aspects of the history that are germane to renal disease include a history of hypertension (which can cause CKD or more commonly be a consequence of CKD), diabetes mellitus, abnormal urinalyses, and problems with pregnancy such as preeclampsia or early pregnancy loss. A careful drug history should be elicited: patients may not volunteer use of analgesics, for example. Other drugs to consider include nonsteroidal anti-inflammatory agents, gold, penicillamine, antimicrobials, chemotherapeutic agents, antiretroviral agents, proton pump inhibitors, phosphate-containing bowel cathartics, and lithium, as well as prior exposure to medical imaging radiocontrast agents. In evaluating the uremic syndrome, questions about appetite, weight loss, nausea, hiccups, peripheral edema, muscle cramps, pruritus, and restless legs are especially helpful. A careful family history of kidney disease, together with assessment of manifestations in other organ systems such as auditory, visual, integumentary and others, may lead to the diagnosis of a heritable form of CKD (e.g., Alport or Fabry syndrome, cystinuria, among others) or shared environmental exposure to nephrotoxic agents (e.g., heavy metals, aristolochic acid). It should be noted that clustering of CKD, sometimes of different etiologies, is often observed within families.

Physical examination

The physical examination should focus on blood pressure and target organ damage from hypertension. Thus, funduscopy and precordial

examination (left ventricular heave, a fourth heart sound) should be carried out. Funduscopy is important in the diabetic patient, as it may show evidence of diabetic retinopathy, which is associated with nephropathy. Other physical examination manifestations of CKD include edema and sensory polyneuropathy. The finding of asterixis or a pericardial friction rub not attributable to other causes usually signifies the presence of the uremic syndrome.

Ancillary Investigations

Laboratory studies should focus on a search for clues to an underlying causative or aggravating disease process and on the degree of renal damage and its consequences. Serum and urine protein electrophoresis, looking for multiple myeloma, should be obtained in all patients >35 years with unexplained CKD, especially if there is associated anemia and elevated, or even inappropriately normal, serum calcium concentration in the face of renal insufficiency. In the presence of glomerulonephritis, autoimmune diseases such as lupus and underlying infectious etiologies such as hepatitis B and C and HIV should be assessed. Serial measurements of renal function should be obtained to determine the pace of renal deterioration and ensure that the disease is truly chronic rather than acute or subacute and hence potentially reversible. Serum concentrations of calcium, phosphorus, vitamin D, and PTH should be measured to evaluate metabolic bone disease. Hemoglobin concentration, iron, B12, and folate should also be evaluated. A 24-h urine collection may be helpful, as protein excretion >300 mg may be an indication for therapy with ACE inhibitors or ARBs.

Imaging Studies

The most useful imaging study is a renal ultrasound, which can verify the presence of two kidneys, determine if they are symmetric, provide an estimate of kidney size, and rule out renal masses and evidence of obstruction. The diagnosis

of renovascular disease can be undertaken with different techniques, including Doppler sonography, nuclear medicine studies, or CT or MRI studies. If there is a suspicion of reflux nephropathy (recurrent childhood urinary tract infection, asymmetric renal size with scars on the renal poles), a voiding cystogram may be indicated. However, in most cases by the time the patient has CKD, the reflux has resolved, and even if still present, repair does not improve renal function. Radiographic contrast imaging studies are not particularly helpful in the investigation of CKD. Intravenous or intra arterial dye should be avoided where possible in the CKD patient, especially with diabetic nephropathy, because of the risk of radiographic contrast dye-induced renal failure. When unavoidable, appropriate precautionary measures include avoidance of hypovolemia at the time of contrast exposure, minimization of the dye load, and choice of radiographic contrast preparations with the least nephrotoxic potential. Additional measures thought to attenuate contrast-induced worsening of renal function include judicious administration of sodium bicarbonate-containing solutions and N-acetyl-cysteine.

Renal Biopsy

In the patient with bilaterally small kidneys, renal biopsy is not advised because (1) it is technically difficult and has a greater likelihood of causing bleeding and other adverse consequences, (2) there is usually so much scarring that the underlying disease may not be apparent, and (3) the window of opportunity to render disease-specific therapy has passed. Other contraindications to renal biopsy include uncontrolled hypertension, active urinary tract infection, bleeding diathesis (including ongoing anticoagulation), and severe obesity. Ultrasound-guided percutaneous biopsy is the favored approach, but a surgical or laparoscopic approach can be considered, especially in the patient with a single kidney where direct visualization and control of bleeding are crucial. In the CKD patient in whom a kidney biopsy is

indicated (e.g., suspicion of a concomitant or superimposed active process such as interstitial nephritis or in the face of accelerated loss of GFR), the bleeding time should be measured, and, if increased, desmopressin should be administered immediately prior to the procedure.

MCQs

- Which of the following statements is accurate regarding the presentation of nephrotic syndrome
 - Hematuria is among the most common presenting symptoms in adults with nephrotic syndrome
 - The first sign of nephrotic syndrome in children is usually swelling of the face
 - The presence of deep venous thrombosis or pulmonary embolism suggests a diagnosis other than nephrotic syndrome
 - Weight loss and hypotension are frequently present in adults with nephrotic syndrome
- Which of the following is accurate regarding BPH and prostate-related issues?
 - All men with BPH demonstrate lower urinary tract symptoms (LUTS), and all men with LUTS have BPH
 - The risk for acute urinary retention (AUR) increases with age
 - Prostate volume typically decreases over time in men with BPH
 - Complications (eg, bladder stones or recurrent urinary tract infections [UTIs]) are frequent
- Of the following, which is more commonly recognized as a secondary cause of nephrotic syndrome
 - Sjögren syndrome
 - Cushing disease
 - Hemolytic anemia
 - Amyloidosis
- In distinguishing acute from chronic renal failure, which of the following is the most reliable evidence of CKD?
 - History of increased serum creatinine

- b. History of increased BUN
 - c. Above-normal plasma sodium
 - d. Above-normal serum calcium
5. Which statement regarding the diagnosis of CKD is most accurate?
- a. Normochromic normocytic anemia typically begins in late-stage CKD
 - b. Measurement of GFR alone is sufficient for identifying stage 1 and stage 2 CKD
 - c. Hypertension is typically a marker of CKD
 - d. eGFR lacks precision and accuracy until GFR reaches $< 60 \text{ mL/min/1.73 m}^2$
6. The severity of BPH assessment is done by
- a. International Prostate Symptom Score (IPSS)
 - b. American nephrological Association Symptom Index (ANA-SI)
 - c. Life quality scale
 - d. Quality of life (QOL)

Answer

1.	b	2.	b	3.	d
4.	a	5.	d	6.	a